

dongwookwon

ai engineer • agentic ai systems

contact

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Seoul, Republic of
Korea

dongwooky.github.io
github / dongwooky
in / dongwooky

languages

korean (native)
english (fluent)

skills

- Python, C/C++
- PyTorch, Transformers
- LangChain, LLMs
- Linux, Docker, CUDA
- GNNs, time-series
- biomedical signals

interests

multi-agent LLMs
anomaly detection
nuclear cybersecurity
amateur radio

education

- 2024–2026 **M.S.** in Computer Engineering, Kwangwoon University, Seoul
GPA 4.33 / 4.5 (98.1%). Bio-Computing & Machine Learning Laboratory.
- 2025 **Graduate Research Program**, University of Toronto
Mechanical and Industrial Engineering. GPA 3.68 / 4.0.
- 2019–2024 **B.S.** in Electronics and Communications Engineering, Kwangwoon University, Seoul
GPA 3.41 / 4.5 (88.1%).

experience

- 2025–Now **SURESOFT TECHNOLOGIES**, Seongnam, Korea
AI Engineer, Intelligent Agent Technology Team. Developing an AI validation system using agentic AI architectures. Designing autonomous and collaborative agents for systematic, scalable AI verification.
- 2021–2025 **BIO-COMPUTING & ML LAB, KWANGWOON U.**, Seoul, Korea
Research Assistant. Deep-learning for anomaly detection in time-series data (ECG, cyber-physical systems). Co-developed an Active Kill-Switch and Biomarker-Based Defense System for IoT medical devices.
- 2025 **LG ELECTRONICS CANADA**, Toronto, ON, Canada
Visiting Researcher (with the University of Toronto). Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- 2022 **QUALCOMM INSTITUTE, UC SAN DIEGO**, San Diego, CA
AI Development Project, Summer Program. Personality-based drug-addiction prediction system. Received the program's Achievement Award.

publications

- 2026 **CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades**, ACL 2026 – Industry Track (Poster)
R. Chang[†], **D. Kwon[†]**, J. Lee[†], N. Verma. [†]Equal contribution.
- 2025 **GEMMAS: Graph-based Evaluation Metrics for Multi-Agent Systems**, EMNLP 2025 – Industry Track (Oral)
J. Lee[†], R. Chang[†], **D. Kwon[†]**, H. Singh, N. Verma. [†]Equal contribution.
- 2025 **Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation**, Data Mining & Knowledge Discovery (under review)
D. Kwon et al.
- 2024 **Hybrid GAT + Transformer for implantable-device anomaly detection**, IBEC 2024 – Best Poster Silver
D. Kwon et al.

awards

- 2022 **Achievement Award** , *Qualcomm Institute AI Project – UC San Diego*
Honored for project work during the summer program.
- 2021 **Gold Award (Ministerial)** , *Smart Maritime Logistics Project – Minister of Oceans and Fisheries*
Top award for project on container detection and stack-layer identification.
- 2025 **Silver Award** , *23rd Embedded Software Competition (KIISE)*
Recognition at one of Korea's flagship embedded-software competitions.

patents

- 2026 **Method for Implantable Medical Device to Detect Anomaly in Real Time** , *KR 10-2961984*
Issued patent, Republic of Korea. April 30, 2026.
- 2025 **GPT API-Based Network Anomaly Detection** , *KR 10-2848814*
Issued patent, Republic of Korea. August 18, 2025.